Chapter 1
Probability and Inference

STAT 5644 - Dr. Ommen

Week 1, Fall 2026

Outline

- Review of Probability
- What is probability?
- Bayesian Statistics
- Why or why not Bayesian?

Events

Definition

The set, Ω, of all possible outcomes of a particular experiment is called the sample space for the experiment.

Definition

An event is any collection of possible outcomes of an experiment, that is, any subset of Ω (including Ω itself).

Example

Craps:

- A dice game where players make bets on the outcome of rolling a pair of dice
  - Come-out roll win: the sum of the dice is 7 or 11
  - Come-out roll loss: the sum of the dice is 2, 3, or 12
  - Come-out roll establishes a point: the sum of the dice is 4, 5, 6, 8, 9, or 10

- Sample Space:
- Events:
  - Come-out roll wins
  - Come-out roll loses
  - Come-out roll establishes a point

Pairwise disjoint

Definition

Two events A_1 and A_2 are disjoint (or mutually exclusive) if both A_1 and A_2 cannot occur simultaneously, i.e. $A_1 \cap A_2 = \emptyset$.

Definition

The events $A_1, A_2, \ldots$ are pairwise disjoint (or pairwise mutually exclusive) if A_i and A_j cannot occur simultaneously for all $i \neq j$, i.e. $A_i \cap A_j = \emptyset$.

Craps pairwise disjoint examples:

Kolmogorov's axioms of probability

Definition

Given a sample space Ω and event space E, a probability is a function $P : E \to \mathbb{R}$ that satisfies

1. $P(A) \geq 0$ for any $A \in E$
2. $P(\Omega) = 1$
3. If $A_1, A_2, \ldots \in E$ are pairwise disjoint, then $P(A_1 \cup A_2 \cup \ldots) = \sum_{i=1}^{\infty} P(A_i)$.

Craps come-out roll probabilities

The following table provides the probability mass function for the sum of the two dice if we *assume* the probability of each elementary outcome is equal:

Outcome	2	3	4	5	6	7	8	9	10	11	12	Sum
Combinations	1	2	3	4	5	6	5	4	3	2	1	36

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The following table provides the probability mass function for the sum of the two dice if we *assume* the probability of each elementary outcome is equal:

Outcome	2	3	4	5	6	7	8	9	10	11	12	Sum
Combinations	1	2	3	4	5	6	5	4	3	2	1	36
Probability	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	1

Craps probability examples:

- P(Win)
- P(Loss)
- P(Point)

Partition

Definition

A set of events, $\{A_1, A_2, \ldots\}$, is a partition of the sample space Ω if and only if

- the events in $\{A_1, A_2, \ldots\}$ are pairwise disjoint and
- $\cup_{i=1}^{\infty} A_i = \Omega$.

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Craps partition examples:

Conditional probability

Definition

If A and B are events in E, and $P(B) > 0$, then the conditional probability of A given B, written $P(A|B)$, is

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

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Example (Craps conditional probability)

$$P(A_7|\text{Win})$$

Law of Total Probability

Corollary (Law of Total Probability)

Let $A_1, A_2, \ldots$ be a partition of Ω and B is another event in Ω. The Law of Total Probability states that

$$P(B) = \sum_{i=1}^{\infty} P(B \cap A_i) = \sum_{i=1}^{\infty} P(B|A_i)P(A_i).$$

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Example (Craps Win Probability)

Let A_i be the event that the sum of two die rolls is i. Then

$$P(\text{Win})$$

Bayes' Rule

Theorem (Bayes' Rule)

If A and B are events in E with $P(B) > 0$, then Bayes' Rule states

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} = \frac{P(B|A)P(A)}{P(B|A)P(A) + P(B|A^c)P(A^c)}$$

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Example (Craps Bayes' Rule)

$$P(A_7|\text{Win})$$

Down Syndrome screening

If a pregnant woman has a test for Down syndrome and it is positive, what is the probability that the child will have Down syndrome?

- Let D indicate a child with Down syndrome and D^c the opposite.
- Let $+$ indicate a positive test result and $-$ a negative result.

$$\begin{aligned} \text{sensitivity} &= P(+|D) &= 0.94 \\ \text{specificity} &= P(-|D^c) &= 0.77 \\ \text{prevalence} &= P(D) &= 0.001 \end{aligned}$$

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What is probability?

Consider the following three typical uses of the word "probability":

- What is the probability I will win on the come-out roll in craps?
- What is the probability my unborn child has Down's syndrome given that they tested positive in an initial screening?
- What is the probability the San Francisco 49ers will win this year's Superbowl?

Win on the come-out roll in craps

To win on the come-out roll in craps requires that the sum of two fair six-sided die is either a 7 or an 11. We calculated this probability earlier (based on equal probabilities of all simple outcomes) to be $2/9$. We likely meant that if we were to repeatedly roll the die, the long term proportion of wins (7s and 11s) would be $2/9$:

Win on the come-out roll in craps

Definition

The frequency interpretation of probability is based on the relative frequency of an event (assumed to be performed in an identical manner).

Two problems with this frequency interpretation:

Down's syndrome

What is the probability my unborn child has Down's syndrome given that they tested positive in an initial screening?

Down's syndrome

What is the probability my unborn child has Down's syndrome given that they tested positive in an initial screening?

Here the frequency interpretation makes no sense for two reasons:

- There is only one child and thus no repeat of the experiment.
- There is no randomness: either the child has Down's syndrome or does not.

Instead, we only have our own uncertainty about whether the child has Down's syndrome.

Down's syndrome

Also, why are we only conditioning on the positive test result, shouldn't we condition on everything else that is important, e.g. age. Then the probability we care about is

$$P(D|+, \text{mother is 33}) = \frac{P(+|D, \text{mother is 33})P(D|\text{mother is 33})}{P(+|\text{mother is 33})}$$

Now the specificity, sensitivity, and prevalence are all the relative frequency of the event for this subpopulation.

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- But what about other measured variables? (e.g. Caucasian, lives in MN, of Scandanavian descent, etc.)
- Taken to its logical extreme, each probability becomes a statement about one single event (e.g. for this individual.)

Superbowl Champions

What is the probability the San Francisco 49ers win the Superbowl?

Superbowl Champions

What is the probability the San Francisco 49ers win the Superbowl?

By similar arguments:

- There is only one Superbowl this year and only one San Francisco 49ers.
- Is the world random? (i.e. Do we have free will?)
    - If not, then (with enough time, computing power, money, etc) we could model the world and know what the result will be.
    - If yes, is there an objective probability that we could be estimating?

Personal belief

Definition

A subjective probability describes an individual's personal judgement about how likely a particular event is to occur.

http://www.stats.gla.ac.uk/glossary/?q=node/488

Remark Coherence of bets. The probability p you assign to an event E is the fraction at which you would exchange p for a return of 1 if E occurs.

Rational individuals can differ about the probability of an event by having different knowledge, i.e. $P(E|K_1) \neq P(E|K_2)$. But given enough data, we might have $P(E|K_1, y) \approx P(E|K_2, y)$.

Personal belief

Using a personal belief definition of probability, it is easy to reconcile the use of probability in common language:

- What is the probability I will win on the come-out roll in craps?
- What is the probability my unborn child has Down's syndrome given that they tested positive in an initial screening?
- What is the probability the San Francisco 49ers will win this year's Superbowl?
- What is the probability that global climate change is primarily driven by human activity?
- What is the probability the Higgs Boson exists?

and in the mathematical notation:

- $p(\theta) \to p(\theta|y)$
- $p(H_1) \to p(H_1|y)$
- $p(\tilde{y}) \to p(\tilde{y}|y)$

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A Bayesian statistician

Let

- y be the data we will collect from an experiment,
- K be everything we know for certain about the world (aside from y), and
- θ be anything we don't know for certain.

One possible definition of a Bayesian statistician is an individual who makes decisions based on the probability distribution of those things we don't know conditional on what we know, i.e.

$$p(\theta|y, K).$$

Bayes' Rule

Bayes' Rule applied to a partition $P = \{A_1, A_2, \ldots\}$,

$$P(A_i|B) = \frac{P(B|A_i)P(A_i)}{P(B)} = \frac{P(B|A_i)P(A_i)}{\sum_{i=1}^{\infty} P(B|A_i)P(A_i)}$$

Bayes' Rule also applies to probability density (or mass) functions, e.g.

$$p(\theta|y) = \frac{p(y|\theta)p(\theta)}{p(y)} = \frac{p(y|\theta)p(\theta)}{\int p(y|\theta)p(\theta)d\theta}$$

where the integral plays the role of the sum in the previous statement.

Bayesian Essentials

Let y be data from some model with unknown parameter θ. Then

$$p(\theta|y) = \frac{p(y|\theta)p(\theta)}{p(y)} = \frac{p(y|\theta)p(\theta)}{\int p(y|\theta)p(\theta)d\theta}$$

and we use the following terminology

Terminology	Notation
Posterior	$p(\theta\|y)$
Prior	$p(\theta)$
Model (likelihood)	$p(y\|\theta)$
Prior predictive (marginal)	$p(y)$

Example: Exponential model

Complete Activity #1

Fun with Bayesian Essentials
Part 1

Let $Y|\theta \sim Exp(\theta)$, then this defines the model density, i.e.

$$p(y|\theta) = \theta e^{-\theta y}.$$

Let's assume a convenient prior $\theta \sim Gamma(a, b)$, then

$$p(\theta) = \frac{b^a}{\Gamma(a)} \theta^{a-1} e^{-b\theta}.$$

Example: Exponential model

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The marginal density (A.K.A prior predictive) is

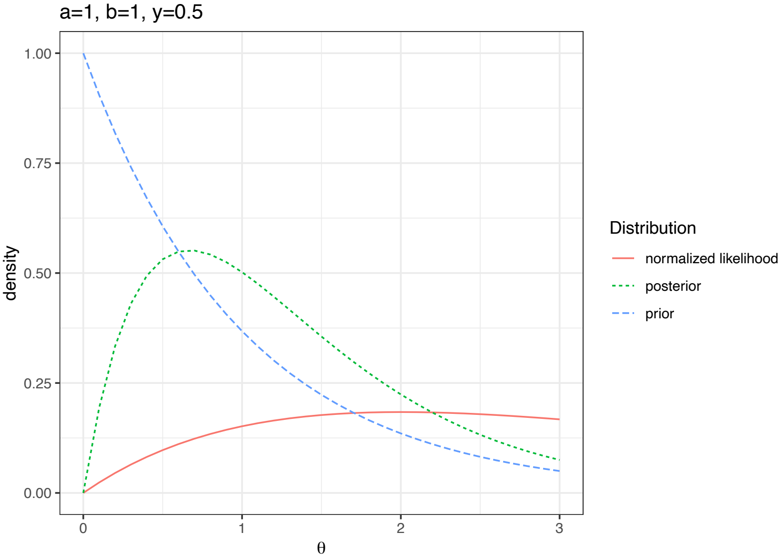
$$p(y) = \int p(y|\theta)p(\theta)d\theta = \frac{b^a}{\Gamma(a)} \frac{\Gamma(a+1)}{(b+y)^{a+1}}.$$

The posterior is

$$p(\theta|y) = \frac{p(y|\theta)p(\theta)}{p(y)} = \frac{(b+y)^{a+1}}{\Gamma(a+1)} \theta^{a+1-1} e^{-(b+y)\theta},$$

thus $\theta|y \sim Gamma(a+1, b+y)$.

Example: Exponential model



Example: Exponential model

Complete Activity #2

Fun with Bayesian Essentials
Part 2

Suppose $Y_i|\theta \overset{iid}{\sim} Exp(\theta)$ for $i = 1, \ldots, n$ and $y = (y_1, \ldots, y_n)$, then the likelihood function is given by

$$p(y|\theta) = \prod_{i=1}^{n} p(y_i|\theta) = \theta^n e^{-\theta n \overline{y}}$$

Let's assume a convenient prior $\theta \sim Gamma(a, b)$, then

$$p(\theta) = \frac{b^a}{\Gamma(a)} \theta^{a-1} e^{-b\theta}.$$

Example: Independent data from Exponential model

Suppose $Y_i|\theta \overset{iid}{\sim} Exp(\theta)$ for $i = 1, \ldots, n$ and $y = (y_1, \ldots, y_n)$, then the likelihood function is given by

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The marginal likelihood (A.K.A prior predictive) is

$$p(y) = \int p(y|\theta) p(\theta) d\theta = \frac{b^a}{\Gamma(a)} \frac{\Gamma(a+n)}{(b + n\overline{y})^{a+n}}.$$

The posterior is

$$p(\theta|y) = \frac{p(y|\theta)p(\theta)}{p(y)} = \frac{(b + n\overline{y})^{a+n}}{\Gamma(a+n)} \theta^{(a+n)-1} e^{-(b+n\overline{y})\theta},$$

thus $\theta|y \sim Gamma(a + n, b + n\overline{y})$.

A shortcut

If

$$p(y) = \int p(y|\theta)p(\theta)d\theta < \infty,$$

then we can actually use the following to find the posterior

$$p(\theta|y) \propto p(y|\theta)p(\theta)$$

where the $\propto$ signifies that terms not involving θ (or anything on the left of the conditioning bar) are irrelevant and can be dropped.

Definition

The kernel of a probability density (mass) function is the form of the pdf (pmf) with any terms not involving the random variable omitted.

A shortcut

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then we can actually use the following to find the posterior

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where the $\propto$ signifies that terms not involving θ (or anything on the left of the conditioning bar) are irrelevant and can be dropped.

In the exponential example

$$p(\theta|y) \propto p(y|\theta)p(\theta) \propto \theta e^{-\theta y}\theta^{a-1}e^{-b\theta} = \theta^{a+1-1}e^{-(b+y)\theta}$$

where we can recognize $p(\theta|y)$ as the kernel of a $Gamma(a+1, b+y)$ distribution and thus $\theta|y \sim Gamma(a+1, b+y)$ and $p(y) < \infty$.

Independent data

Suppose $Y_i|\theta \stackrel{ind}{\sim} Exp(\theta)$ for $i = 1, \ldots, n$ and $y = (y_1, \ldots, y_n)$, then

$$p(y|\theta) = \prod_{i=1}^{n} p(y_i|\theta) = \theta^n e^{-\theta n\overline{y}}$$

Let's assume a convenient prior $\theta \sim Gamma(a, b)$, then

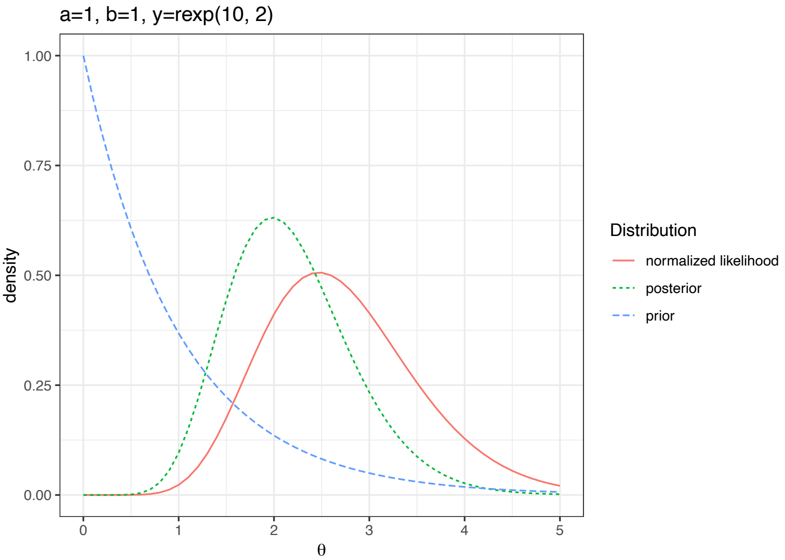
$$p(\theta) = \frac{b^a}{\Gamma(a)}\theta^{a-1}e^{-b\theta}$$

Then

$$p(\theta|y) \propto p(y|\theta)p(\theta) \propto \theta^{a+n-1}e^{-(b+n\overline{y})\theta}$$

We recognize this as the kernel of a gamma, i.e. $\theta|y \sim Gamma(a + n, b + n\overline{y})$.

Example: Independent data from Exponential model



Bayesian Parameter Estimation

Goal: Base estimation off of the posterior distribution for that parameter

- Explicit conditioning:

  $$p(\theta|y, M)$$

  where M is a model with parameter (vector) θ and y is data assumed to come from model M with true parameter θ_0.

- Implicit conditioning:

  $$p(\theta|y)$$

  where θ is the unknown parameter (vector) and y is the data.

Bayesian learning (in parameter estimation)

So, Bayes' Rule provides a formula for updating from prior beliefs to our posterior beliefs based on the data we observe, i.e.

$$p(\theta|y) = \frac{p(y|\theta)}{p(y)} p(\theta) \propto p(y|\theta)p(\theta)$$

Suppose we gather $y_1, \ldots, y_n$ sequentially (and we assume y_i independent conditional on θ), then we have

$$\begin{aligned} p(\theta|y_1) &\propto p(y_1|\theta)p(\theta) \\ p(\theta|y_1, y_2) &\propto p(y_2|\theta)p(\theta|y_1) \end{aligned}$$

and

$$p(\theta|y_1, \ldots, y_i) \propto p(y_i|\theta)p(\theta|y_1, \ldots, y_{i-1})$$

So Bayesian learning is

$$p(\theta) \to p(\theta|y_1) \to p(\theta|y_1, y_2) \to \cdots \to p(\theta|y_1, \ldots, y_n).$$

Bayesian Hypothesis Testing/Model Comparison

Goal: Base testing/comparison on the posterior distribution for that hypothesis/model

- Explicit conditioning:

$$p(M_j|y, \mathcal{M})$$

  where $\mathcal{M}$ is a set of models with $M_j \in \mathcal{M}$ for $i = 1, 2, \ldots$ and y is data assumed to come from some model $M_0 \in \mathcal{M}$.

- Implicit conditioning:

$$p(M_j|y)$$

  where M_j is one of a set of models under consideration and y is data assumed to come from one of those models.

Bayesian Model Comparison

Formally, to compare models (or average over models), we use

$$p(M_j|y) \propto p(y|M_j)p(M_j)$$

where

- $p(y|M_j)$ is the likelihood of the data when model M_j is true
- $p(M_j)$ is the prior probability for model M_j
- $p(M_j|y)$ is the posterior probability for model M_j

Thus, a Bayesian approach provides a natural way to learn about models, i.e. $p(M_j) \rightarrow p(M_j|y)$.

Bayesian Prediction

Let y be observed data and $\tilde{y}$ be unobserved data from a model with parameter θ where $\tilde{y}$ is conditionally independent of y given θ
(true for many of the models we will discuss this semester),
then the **posterior predictive** distribution is

$$\begin{aligned} p(\tilde{y}|y) &= \int p(\tilde{y},\theta|y)d\theta \\ &= \int p(\tilde{y}|\theta,y)p(\theta|y)d\theta \\ &= \int p(\tilde{y}|\theta)p(\theta|y)d\theta \end{aligned}$$

where $p(\theta|y)$ is the posterior we obtained using Bayesian parameter estimation techniques.

Example: Exponential model

Complete Activity #3

Fun with Bayesian Essentials
Part 3

From the previous example, let $y_i \overset{iid}{\sim} Exp(\theta)$ and $\theta \sim Gamma(a, b)$, then $\theta|y \sim Gamma(a+n, b+n\overline{y})$. Suppose we are interested in predicting a new value $\tilde{y} \sim Exp(\theta)$ (conditionally independent of $y = (y_1, \ldots, y_n)$ given θ).

Example: Exponential model

From the previous example, let $y_i \overset{iid}{\sim} Exp(\theta)$ and $\theta \sim Gamma(a, b)$, then $\theta|y \sim Gamma(a+n, b+n\overline{y})$. Suppose we are interested in predicting a new value $\tilde{y} \sim Exp(\theta)$ (conditionally independent of $y = (y_1, \ldots, y_n)$ given θ). Then we have

$$\begin{aligned}
p(\tilde{y}|y) &= \int p(\tilde{y}|\theta)p(\theta|y)d\theta \\
&= \int \theta e^{-\theta\tilde{y}} \frac{(b+n\overline{y})^{a+n}}{\Gamma(a+n)} \theta^{a+n-1} e^{-\theta(b+n\overline{y})} d\theta \\
&= \frac{(b+n\overline{y})^{a+n}}{\Gamma(a+n)} \int \theta^{a+n+1-1} e^{-\theta(b+n\overline{y}+\tilde{y})} d\theta \\
&= \frac{(b+n\overline{y})^{a+n}}{\Gamma(a+n)} \frac{\Gamma(a+n+1)}{(b+n\overline{y}+\tilde{y})^{a+n+1}} \\
&= \frac{(a+n)(b+n\overline{y})^{a+n}}{(\tilde{y}+b+n\overline{y})^{a+n+1}}
\end{aligned}$$

This is the Lomax distribution for $\tilde{y}$ with parameters $a+n$ and $b+n\overline{y}$.

Bayesian Prediction

Goal: Base prediction on the posterior predictive distribution

- Explicit conditioning:
  $$p(\tilde{y}|y, M)$$
  where $\tilde{y}$ is unobserved data and y and $\tilde{y}$ are both assumed to come from M. Alternatively,
  $$p(\tilde{y}|y, \mathcal{M})$$
  where y and $\tilde{y}$ are both assumed to come from some $M_0 \in \mathcal{M}$.
- Implicit conditioning:
  $$p(\tilde{y}|y)$$
  where $\tilde{y}$ is unobserved data and y and $\tilde{y}$ are both assumed to come from the same (set of) model(s).

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Why or why not Bayesian?

Why do a Bayesian analysis?

- Incorporate prior knowledge via $p(\theta)$
- Coherent, i.e. everything follows from specifying $p(\theta|y)$
- Interpretability of results, e.g. the probability the parameter is in (L, U) is 95%

Why not do a Bayesian analysis?

- Need to specify $p(\theta)$
- Computational cost
- Does not guarantee coverage, i.e. how well do the procedures work over all their uses (although *frequentist matching* priors are specifically designed to ensure frequentist properties, e.g. coverage)